

Research Article

Blended Use of Corn Cob and Rice Husk Ashes in Sustainable Mortar Production

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Abstract

The use of agricultural and industrial by-products in sustainable material production has become an increasingly attractive strategy for waste management with growing environmental awareness about the possible adverse effects of waste materials. Therefore, the partial replacement of cement by mass with corn cob ash (CCA) and rice husk ash (RHA) to develop sustainable mortar composites was investigated in this study. A three-phase experimental program was conducted, including (i) pozzolanic activity assessment, (ii) optimization of CCA–RHA mix ratio, and (iii) evaluation of mechanical and durability properties at various blended replacement levels (5–20% by mass). Key tests such as compressive and flexural strength, XRD, TGA, and SEM showed enhanced cementitious performance with 5–10% cement replacement. RHA accelerated early pozzolanic reactions, while CCA contributed to long-term durability. Microstructural analyses indicated reduced calcium hydroxide (CH) content and increased calcium silicate hydrate (C-S-H) formation. Results show that blended agricultural ashes offer effective SCM potential and contribute to sustainability by utilizing waste materials.

Keywords: *Supplementary Cementitious Materials (SCM), Agricultural Waste, Corn Cob Ash, Rice Husk Ash, Pozzolanic Activity*

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Sürdürülebilir Harç Üretiminde Mısır Koçanı ve Pirinç Kabuğu Küllerinin Birlikte Kullanımı

Öz

Atık malzemelerin çevre üzerindeki olumsuz etkilerine dair artan farkındalık, tarımsal ve endüstriyel yan ürünlerin sürdürülebilir malzeme üretiminde değerlendirilmesini atık yönetimi açısından giderek daha önemli bir strateji haline getirmiştir. Bu çalışmada, mısır koçanı külü (CCA) ve pirinç kabuğu külü (RHA) çimento ile ağırlıkça kısmi olarak ikame edilerek sürdürülebilir harç üretimi geliştirilmesi araştırılmıştır. Üç aşamalı deneysel yaklaşım kapsamında (i) puzolanik aktivite değerlendirmesi, (ii) CCA-RHA karışım oranının optimizasyonu ve (iii) mekanik ve dayanıklılık performanslarının farklı çimento ikame oranlarında (ağırlıkça 5–20%) incelenmesi gerçekleştirilmiştir. Basınç ve eğilme dayanımı testleri, XRD, TGA ve SEM analizleri ile yapılan değerlendirmelerde, 5–10% çimento ikame oranlarının en uygun performansı sağladığı görülmüştür. RHA erken yaşta puzolanik reaksiyonları hızlandırırken, CCA uzun vadeli dayanıklılığa katkı sağlamıştır. Mikroyapı analizleri, kalsiyum hidroksit (CH) içeriğinde azalma ve kalsiyum silikat hidrat (C-S-H) oluşumunda artış olduğunu göstermiştir. Bulgular, bitkisel kökenli atık küllerinin puzolanik potansiyele sahip olduğunu ve atık malzemelerin değerlendirilmesi yoluyla sürdürülebilirliğe katkı sunduğunu desteklemektedir.

Anahtar Kelimeler: İkincil Bağlayıcı malzeme (SCM), Tarımsal Atık, Mısır Koçanı Külü, Pirinç Kabuğu Külü, Puzolanik Aktivite

1. Introduction

Cement production remains one of the most carbon-intensive industrial processes worldwide, responsible for approximately 5-8% of global anthropogenic CO₂ emissions (Ekinçioğlu et al., 2013), and more recently reported as 6% based on updated global inventories (Tiseo, 2025). Its environmental footprint extends beyond greenhouse gas emissions, encompassing particulate pollution, resource depletion, and significant energy consumption. In response to these challenges, research into supplementary cementitious materials (SCMs) has intensified, aiming to reduce Portland cement usage while maintaining or enhancing the mechanical and durability performance of cementitious systems.

SCMs come from many sources and are grouped into different categories. Early studies evaluated the efficacy of natural pozzolans—such as volcanic ash, diatomaceous earth, and pumice—for their silica-rich composition and long-term durability enhancement. With industrialization, by-products like fly ash (FA), ground granulated blast furnace slag (GGBFS), and silica fume (SF) gained importance due to their high reactivity and resource optimization opportunities, especially in large-scale applications. These materials react with calcium hydroxide (CH) through secondary hydration reactions, forming additional calcium silicate hydrate (C-S-H) gel, which contributes to denser microstructures and improved long-term properties.

Meanwhile, the utilization of waste materials as SCMs has emerged as a promising way for sustainable construction practices, offering both environmental benefits and enhanced performance characteristics. More recently, the use of agricultural waste-based SCMs has gained attention, particularly for their role in sustainability, waste valorization, and local material sourcing. Notable examples include rice husk ash (RHA), corn cob ash (CCA), wheat straw ash (WSA), and bagasse ash (BA). Among them, RHA has been extensively studied due to its high amorphous silica content (often exceeding 80%) and its ability to react rapidly when properly calcined and milled. Prior investigations (e.g., Bahri et al., 2019; Le et al., 2014; Noori et al., 2023; Jayaraman et al., 2023) report consistent enhancements in compressive and flexural strength, durability, and alkali-silica reaction (ASR) resistance at replacement levels of 10–15%.

In contrast to RHA, corn cob ash (CCA) remains relatively less studied. Its chemical reactivity and pozzolanic potential depend strongly on calcination conditions, particle fineness, and agricultural origin. Previous studies have demonstrated moderate silica content and delayed strength development (Kamau et al., 2016; Abubakar et al., 2016), with significant improvements observed only when CCA is calcined at approximately 700 °C and finely ground. Under optimal conditions, its pozzolanic activity index (PAI) (referred to as the strength activity index (SAI) in ASTM C311) can

approach the minimum threshold required for classification as an SCM. Replacement levels between 5% and 15% provide 28-day compressive strengths comparable to or exceeding control mixes, with a 10% replacement offering optimal strength gains and improved ductility. Nevertheless, when used alone, CCA may adversely affect early-age strength and workability due to its relatively low reactivity and high alkali content; moreover, dosages above 15% lead to increased water demand, extended setting times, and, in excess of 20%, often require prolonged curing to achieve design strength (Adebisi et al., 2019).

While both RHA and CCA have individually demonstrated SCM viability, their combined use remains largely unexplored. Most previous investigations have examined rice husk ash (RHA) and corn cob ash (CCA) as supplementary cementitious materials independently, leaving limited understanding of how their combined utilization might balance early- and later-age reactivity while mitigating challenges such as alkali-silica reaction and early-age strength reduction.

A few studies (Kumari et al., 2018; Olaniyi, 2022) suggest that blending these ashes could yield synergistic effects in binder design, leveraging RHA's early-age reactivity and CCA's long-term strength contribution. However, prior research lacks standardized experimental comparisons, optimization of blend ratios, and the comprehensive evaluation of mechanical, durability,

and microstructural performance. This study aims to address that gap through a comprehensive three-phase experimental program.

Phase I evaluates the pozzolanic activity indices of RHA and CCA across varying calcination temperatures. Phase II investigates multiple RHA:CCA blend ratios to determine the most favorable combination for mechanical performance. Phase III applies this optimum blend to mortars with replacement levels of 5%, 10%, 15%, and 20%, assessing fresh state properties (workability, air content), strength development, durability (ASR expansion, capillary absorption), and microstructural changes through XRD, TGA, and SEM analyses.

By improving our understanding of blended agricultural waste pozzolans, this study aims to contribute to the development of cementitious systems that are low in carbon and efficient in resource use, contributing to global sustainability efforts.

2. Materials and Methods

This study investigates the pozzolanic performance of corn cob ash (CCA) and rice husk ash (RHA) in mortar systems through a three-phase experimental program. All procedures were conducted in accordance with relevant ASTM and EN standards.

2.1. Ordinary Portland Cement (OPC)

The cement used was CEM I 42.5N type Portland cement, classified according to

BS EN 197-1, and its density was measured as 3.09 g/cm^3 .

2.2. Fine Aggregate

Natural sand used in the mixtures was obtained from the Sakarya River and sieved through a 2 mm sieve to eliminate oversized particles before mixing. The density was determined as 2.66 g/cm^3 , and particle size distribution conformed to the grading requirements specified in BS EN 196-1.

2.3. Superplasticizer

To compensate for the increased water demand of pozzolanic materials, a polycarboxylate-based superplasticizer (MasterGlenium 51) was used at dosages up to 2% by cement weight, in accordance with BS EN 934-2. The density of the used superplasticizer was measured as 1.1 g/cm^3 .

2.4. Rice Husk Ash (RHA)

RHA was commercially obtained from Erdoğanlar Gıda San. ve Tic. Ltd. Şti., having been previously calcined via industrial processes. Before use, the ash was oven-dried and ground in a ball mill to achieve a fineness below $160 \text{ }\mu\text{m}$, suitable for mortar production. The RHA used in the study is shown in Figure 1, and its density was measured as 2.57 g/cm^3 .

2.5. Corn Cob Ash (CCA)

CCA was used after burning and grinding entirely in the laboratory. Corn cobs collected from Sakarya Province as

agricultural waste material were calcined at four different temperatures (400°C , 500°C , 600°C , and 700°C) for 2 hours in a muffle furnace to determine the optimum burning condition. Then, the obtained ash was rapidly cooled with water to promote the formation of an amorphous structure. After cooling, samples were oven-dried at 150°C for 48 hours and subsequently ground using a ball mill to obtain particles finer than $160 \text{ }\mu\text{m}$. The used CCA is shown in Figure 2, and its density was measured as 2.40 g/cm^3 .



Figure 1. Rice husk ash used in the study.



Figure 2. Corn cob ash after burning and grinding.

2.6. Chemical Characterization

The chemical compositions of OPC, RHA, and CCA were determined by X-

ray fluorescence (XRF) analysis, and the oxide contents are presented in Table 1.

The high SiO₂ content in RHA (84.88%) and moderate levels in CCA (25.17%) suggest differing pozzolanic potentials. The combined content of SiO₂, Al₂O₃, and Fe₂O₃ (S+A+F) exceeds 85% for RHA, indicating strong pozzolanic activity, while CCA shows a value slightly higher than 30%, suggesting limited standalone reactivity. However, Olafisu et al. (2018) reported that the S+A+F content of CCA tends to increase with higher calcination temperatures, supporting its potential when thermally optimized. This supports the hypothesis that RHA promotes early calcium silicate hydrate (C–S–H) formation, while CCA contributes to later-age hydrates, although at a slower rate.

Although the raw CCA shows a relatively high loss on ignition (LOI=11.29%), this is expected due to residual organics and moisture. LOI values are known to decrease significantly upon incineration at elevated temperatures, which enhances the ash's reactivity and stability. Therefore, CCA in this study was considered as a supplementary material only after calcination, particularly in blended systems with RHA, where it may contribute to long-term matrix refinement and promote sustainable material utilization.

Low CaO levels in both ashes (1.05% in RHA, 2.45% in CCA) confirm their non-cementitious nature, reinforcing their role as supplementary cementitious materials. The high K₂O content in CCA

(39.70%) may influence alkali–silica reaction (ASR) risk, while low SO₃ levels in both ashes minimize concerns about sulfate-related expansion.

These compositional characteristics form the basis for selecting and interpreting subsequent tests, including SAI, ASR, capillary absorption, microstructural analysis, and mechanical strengths.

Table 1. Major oxides of raw materials determined by XRF analysis.

Compound	OPC (%)	RHA (%)	CCA (%)
SiO ₂	16.83	84.88	25.17
Al ₂ O ₃	3.84	0.22	3.20
Fe ₂ O ₃	3.70	0.28	2.10
MgO	1.06	0.52	3.88
CaO	64.51	1.05	2.45
Na ₂ O	0.32	0.55	0.45
K ₂ O	1.10	3.60	39.70
TiO ₂	0.25	0.31	1.49
P ₂ O ₅	0.20	0.86	5.70
Cr ₂ O ₃	0.12	<0.01	<0.01
SO ₃	4.73	0.31	1.49
LOI	3.07	7.30	11.29

2.7. Experimental Design Overview

The experimental program was conducted in three distinct phases, and Figure 3 provides an overview of the experimental methodology and key steps.

2.7.1. Phase I – Evaluation of Pozzolanic Activity Index (PAI)

In this phase, 20% of cement was replaced with either RHA or CCA calcined at different temperatures.

Compressive strength development at 7 and 28 days was assessed, and PAI values were calculated by comparing the test results with reference samples. A

minimum PAI of 0.75 was adopted as the acceptance threshold, in accordance with BS EN 450-1 and ASTM C618.

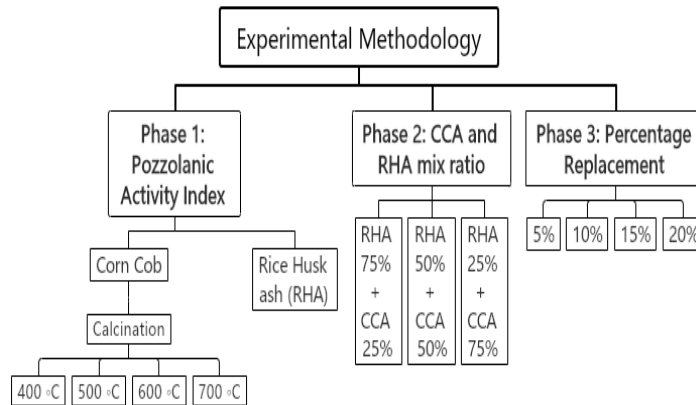


Figure 3. Overview of the experimental methodology.

2.7.2. Phase II –Optimization of RHA and CCA Blend Ratio

In the second phase of the investigation, the focus shifted from individual evaluation to exploring the combined effect of CCA and RHA. For this purpose, CCA with the highest PAI (calcined at 700 °C) was blended with RHA at varying internal ratios while maintaining a total binder replacement level of 20%. Mix designs included 0C20R, 5C15R, 10C10R, 15C5R, and 20C0R. Here, “C” and “R” represent CCA and RHA, respectively, and the preceding numbers indicate their replacement percentages. The water-to-binder ratio was fixed at 0.50. Mortar prisms (40×40×160 mm) were tested for flexural strength via three-point bending at 7 and 28 days in accordance with BS EN 196-1. Compressive strength was measured on the broken halves of mortars after flexural testing. All flexural and compressive strength values

represent the average of three specimens per mix design, tested to ensure consistency and reduce variability in reported results.

2.7.3. Phase III – Effect of Total Replacement Level Using Optimal Blend

In the final phase, the optimal blend ratio identified in phase II (75% CCA + 25% RHA) was used to evaluate the effect of varying total binder replacement levels: 5%, 10%, 15%, and 20%. Mixtures were coded as M5 to M20, corresponding to the total replacement percentage. For example, in M5, 5% of cement was replaced with the optimal blend, comprising 3.75% CCA and 1.25% RHA. The mix designs for the third phase are shown in Table 2.

Fresh-state tests included workability (BS EN 1015-3) and air content (ASTM C185). Hardened-state assessments

involved flexural and compressive strength (BS EN 196-1), capillary water absorption (ASTM C1585, BS EN 1015-18), and ASR expansion (ASTM C1567).

Table 2. Mix design for phase III (optimum replacement percentages).

Materials and ratios	Density (g/cm ³)	M0 (REF)	M5	M10	M15	M20
Water	1	225	225	225	225	225
Sand	2.66	1350	1350	1350	1350	1350
OPC	3.14	450	427.5	405	382.5	360
CCA	2.47	0	16.9	33.75	50.6	67.5
RHA	2.57	0	5.6	11.25	16.9	22.5
S.P.	1.09	6	6	6	6	6
W/B		0.5	0.5	0.5	0.5	0.5
Poz/Binder		0	0.05	0.1	0.15	0.2

2.8. Microstructural Analysis

Microstructural characterization employed the following methods to investigate the pozzolanic reaction mechanisms, including calcium hydroxide (CH) consumption, calcium silicate hydrate (C-S-H) formation, and quartz formation as a function of increasing cement replacement with CCA and RHA.

2.8.1. X-ray Diffraction (XRD)

Five mortar mixtures (M0, M5, M10, M15, and M20) were analyzed at the Prof. Dr. Adnan Tekin Material Sciences and Production Technologies Application Research Center (ATUM) at Istanbul Technical University. Specimens were ground to powder, and the diffraction data were collected in the 10–60° 2 θ range, where the most resolved peaks were observed. Quantitative phase analyses were performed using computer software

(Malvern Panalytical ver. 5.2) to assess the evolution of hydration products.

2.8.2. Thermogravimetric Analysis (TGA)

Thermogravimetric analysis (TGA) was performed on the same five mixtures (M0-M20) to quantify calcium hydroxide content through decomposition profiles. The samples were ground to a fine powder and placed in ceramic crucibles. Heating was performed from room temperature to 650 °C at a rate of 10 °C/min under static air conditions. Weight loss between 400–600°C was used to estimate CH decomposition, providing insight into the extent of pozzolanic reaction.

2.8.3. Scanning Electron Microscopy (SEM)

SEM imaging was carried out at the Membrane Technologies Application-Research Center (MEM-TEK), Istanbul Technical University, for five mixtures

(M0-M20). Mortar specimens were crushed into approximately 1 cm * 1 cm fragments, and at least four pieces from each mixture were prepared. In the provided SEM figures, magnification ratios of 50 μm and 100 μm are used, enabling detailed observation of morphological features such as C-S-H gel formation, reduction in CH crystals, and changes in pore structure at different scales. These observations supported the quantitative findings obtained from XRD and TGA analyses. All SEM evaluations were performed on specimens aged approximately 60 days, allowing the visualization of mature hydration products and long-term microstructural development.

3. Results and Discussion

3.1. Results of Phase I – Evaluation of Pozzolan Activity Index (PAI)

The pozzolanic performance of as-received RHA and CCA, calcined at four different temperatures (400°C, 500°C, 600°C, 700°C), was evaluated based on compressive strength measurements of mortars at 7 and 28 days. Each modified mix contained 20% pozzolanic replacement by weight of OPC. The reference mortar (100% OPC was used as binder) served as the control specimen for calculating PAI values. The results are presented in Table 3, showing the compressive strength and corresponding PAI values at both curing ages.

The data clearly show that, all CCA-based mixes exhibited relatively low PAI values, particularly at early ages, with

most failing below the ASTM C618 minimum threshold of 0.75. This indicates limited pozzolanic activity of CCA, especially during the initial hydration period. Among the CCA variants, only the sample calcined at 700°C (CCA700) approached the threshold at both 7 and 28 days (PAI = 0.75 and 0.76, respectively), suggesting improved reactivity due to increased amorphous silica content and reduced residual organics. This trend aligns with findings by Olaleye (2022), who demonstrated that corncob ash calcined at 700°C can serve as a viable pozzolanic material in concrete applications. Additional support is provided by Olafusi et al. (2018), whose characterization of CCA calcined at approximately 625–650°C confirmed its compliance with ASTM C618 criteria and its suitability as a cementitious additive.

In contrast, RHA exhibited higher PAI values (0.89 at 7 days and 0.83 at 28 days), confirming its superior pozzolanic performance, particularly at early ages. This behavior is attributed to its highly reactive amorphous silica phase and fine particle size, as supported by Siddika et al. (2020), who reported that ash fineness and the presence of amorphous silica significantly accelerate early C–S–H formation and CH consumption, thereby enhancing pozzolanic reactivity.

These findings highlight the temperature- and age-dependent nature of CCA's reactivity and support the selection of CCA700 for further investigation in Phase II. Although

mechanical tests were limited to 7 and 28 days, the observed trends suggest evolving pozzolanic behavior over time. The results also confirm the high

pozzolanic potential of RHA, particularly at early ages.

Table 3. Compressive strength and pozzolanic activity index of RHA and CCA-based mortars.

Sample	Calcination Temp (°C)	7-Day Comp. Strength (MPa)	28-Day Comp. Strength (MPa)	PAI (7-Day)	PAI (28-Day)
REF	—	44.6	59.7	1.00	1.00
RHA	Industrial	39.7	49.4	0.89	0.83
CCA400	400	22.0	33.3	0.49	0.56
CCA500	500	24.9	38.3	0.56	0.64
CCA600	600	26.2	40.1	0.59	0.67
CCA700	700	33.3	45.2	0.75	0.76

3.2. Results of Phase II – Optimization of RHA/CCA Blend Ratio

In Phase II, the total replacement level was fixed at 20% by mass, by varying the internal ratio between RHA and CCA: 0C20R, 5C15R, 10C10R, 15C5R, and 20C0R. The objective was to determine the blend that offered the best mechanical performance while maintaining acceptable workability. The reference mix (M0), consisting entirely of OPC as binder, was included for comparison under an identical water-to-binder ratio ($w/b = 0.50$). The mechanical performance results are summarized in Table 4, showing the flexural and compressive strength results at 7 and 28 days for all mixes.

As shown in Table 4, all modified mixes exhibited lower strength values than the reference (M0), reflecting the expected impact of partial replacement of OPC.

Among the alternatives, 0C20R and 15C5R yielded the highest compressive strength at 28 days (49.4 MPa and 49.2 MPa, respectively), with 15C5R also demonstrating consistent early-age performance. Notably, the 20C0R mix, containing only CCA, exhibited the lowest strength values at both ages, suggesting that increasing CCA content beyond 15% may adversely affect mechanical performance.

Flexural strength results followed a similar trend. At 28 days, the blended mixes containing both RHA and CCA—namely 5C15R, 10C10R, and 15C5R—achieved flexural strengths of 8.3, 8.2, and 8.1 MPa, respectively, corresponding to 77%, 76%, and 75% of the reference value. In contrast, the 20C0R mix reached only 6.4 MPa (59% of M0), indicating that RHA plays a key role in enhancing the mechanical synergy of blended pozzolanic systems.

Table 4. Mechanical properties of mortars with varying CCA/RHA blend ratios.

Mix Code	CCA (%)	RHA (%)	7-Day Flex. St. (MPa)	7-Day Comp. Str. (Mpa)	28-Day Flex. St. (MPa)	28-Day Comp. Str. (MPa)
M0	0	0	9.2	46.0	10.8	59.7
0C20R	0	20	7.3	39.7	8.5	49.4
5C15R	5	15	7.1	38.4	8.3	46.5
10C10R	10	10	6.9	39.2	8.2	48.0
15C5R	15	5	6.9	39.6	8.1	49.2
20C0R	20	0	5.6	33.3	6.4	45.2

Moreover, early-age compressive strength data revealed that RHA-containing mixes (5C15R, 10C10R, and 15C5R) reached 83–86% of the reference (M0) strength at 7 days, whereas the CCA-only mix (20C0R) achieved just 73%. This suggests that RHA contributes to accelerated hydration and early strength development, likely due to its high amorphous silica content and fine particle size. These results suggest that formulations with moderate RHA content and controlled CCA inclusion can sustain mechanical integrity when properly proportioned.

Although the differences in 28-day compressive strength among the top-performing mixes (0C20R, 10C10R, and 15C5R) were relatively small—within a range of 1.4 MPa—the 15C5R mix was selected for subsequent investigation. While the 10C10R blend also showed promising results, 15C5R demonstrated a more favorable balance between early-age strength and long-term performance indicators. Additionally, this blend allowed for a higher proportion of CCA, which is less studied in the literature. Therefore, the composition, comprising 75% CCA and 25% RHA within the

replacement fraction, was considered optimal for Phase III, where the impact of total binder replacement percentage on both fresh and hardened properties was systematically evaluated.

3.3. Results of Phase III – Replacement Percentage Study of Optimal Blend

In phase III, the 15C5R blend, previously identified in Phase II as mechanically optimal, was investigated at four total replacement levels: 5%, 10%, 15%, and 20%. A reference mix (M0), containing 100% OPC, served as the control mixture. All mixtures were prepared with a constant water-to-binder ratio of 0.50 to ensure comparability across tests.

3.3.1. Fresh-State Properties

Workability was assessed via flow table tests in accordance with BS EN 1015-3, while air content was measured following ASTM C185. The results are presented in Figure 4.

A consistent trend of decreasing flow diameter was observed with increasing replacement levels, indicating a gradual reduction in workability. This decline is

primarily attributed to the incorporation of pozzolanic materials—particularly rice husk ash (RHA)—which possess high silica content and extremely fine

particle size. These characteristics increase the surface area available for water absorption, thereby elevating water demand and reducing flowability.

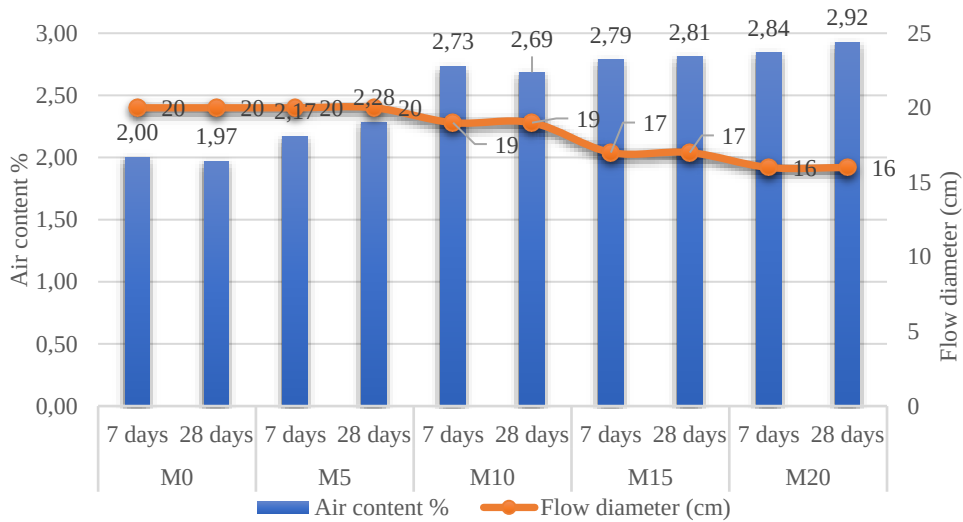


Figure 4. Relation between air content and workability.

Singh (2019) reported that the high surface area and fineness of RHA significantly increase water demand, leading to reduced workability. He recommended the use of chemical admixtures—such as superplasticizers—to restore acceptable flow. More recently, Abellan-García (2023) investigated the fresh properties of concrete containing RHA and likewise attributed the reduction in workability to RHA's high silica content and absorptive nature. These outcomes demonstrate the consistent trend of reduced flow as replacement levels rise.

In the present study, a polycarboxylate-based superplasticizer was employed to mitigate the increased water demand. As a result, flow diameters ranging from 16 cm to 20 cm were achieved at all

replacement levels. These values fall well within the acceptable range for mortar applications, indicating that all mixtures retained satisfactory workability throughout production.

Conversely, air content exhibited a steady increase with rising replacement percentages. This phenomenon is likely due to the irregular morphology and porous nature of RHA and CCA particles, which promote air entrapment during mixing. As shown in Figure 4, an inverse relationship between air content and workability is evident, aligning with established principles in concrete technology: increased air content tends to disrupt the flowability of the mixture, thereby reducing its workability.

To improve workability in mixtures containing agricultural ashes, several

practical strategies may be considered. In this study, a polycarboxylate-based superplasticizer was used to mitigate the elevated water demand associated with the fine and absorptive nature of RHA. The use of blended RHA–CCA systems also helped maintain acceptable flow, likely due to differences in particle morphology and packing behavior, even though both ashes were sieved to the same nominal size before use. Additional approaches—such as

adjusting the water-to-binder ratio within practical limits or applying pre-treatment methods to reduce ash porosity—may further enhance consistency during mixing.

3.3.2. Mechanical Strength

Flexural and compressive strength results at 7 and 28 days are shown in Table 5.

Table 5. Strength development of mortars with increasing replacement ratios.

Mix Code	Replacement (%)	7-Day Flex. St. (MPa)	28-Day Flex. St. (MPa)	7-Day Comp. Str. (MPa)	28-Day Comp. Str. (MPa)
M0	0	9.2	10.9	51.0	61.6
M5	5	8.4	10.1	50.1	59.8
M10	10	7.5	9.9	46.6	56.6
M15	15	7.3	8.7	44.7	52.2
M20	20	6.9	8.1	39.6	49.2

At 28 days, all modified mixes retained between 80% and 97% of the compressive strength of M0, with M5 exhibiting the closest performance to the reference mix. Although a gradual decline in strength was observed with increasing replacement ratios, the reductions remained within acceptable engineering limits. Notably, even the M20 mix, which incorporated the highest replacement level, achieved a compressive strength of 49.2 MPa—approximately 80% of the reference value (61.6 MPa)—indicating its potential suitability for general-purpose mortar applications.

Flexural strength followed a similar downward trend. The reduction in

strength is primarily attributed to the dilution of cementitious content and the relatively moderate pozzolanic activity of CCA. However, the presence of RHA appears to mitigate early-age strength loss, likely due to its high amorphous silica content and micro-filler effect, which enhances particle packing and contributes to early hydration.

These findings suggest that up to 10–15% replacement levels can be implemented without significantly compromising mechanical performance, especially when RHA is included in the blend to offset the limitations of CCA.

3.3.3. Durability Indicators

Two key durability metrics were assessed: the coefficient of capillary water absorption, measured in

accordance with BS EN 1015-18, and alkali-silica reaction (ASR) expansion at 14 days, evaluated following ASTM C1567. The results are summarized in Table 6.

Table 6. Durability performance of mortars.

Mix Code	Replacement (%)	Coef. of Water Absorption (kg/m ² ·min ^{0.5})	ASR Expansion at 14 days (%)
M0	0	0.053	0.34
M5	5	0.057	0.33
M10	10	0.067	0.32
M15	15	0.063	0.30
M20	20	0.080	0.28

Capillary water absorption values showed a gradual increase with higher replacement levels, rising from 0.053 kg/m²·min^{0.5} in the control mix (M0) to 0.080 kg/m²·min^{0.5} at 20% replacement. This trend suggests that the incorporation of CCA and RHA may lead to a more porous matrix, likely due to the reduced cement content and incomplete hydration.

ASR expansion values decreased modestly with increasing replacement, from 0.34% in the control mix to 0.28% at 20% replacement. Although these values remain above the ASTM C1567 threshold of 0.10% for harmless behavior, the downward trend suggests a partial mitigation effect. This reduction is primarily attributed to the high silica content and pozzolanic activity of RHA, which consumes calcium hydroxide and contributes to the formation of additional calcium-silicate-hydrate (C-S-H), thereby densifying the matrix and limiting alkali mobility. Additionally,

the alkali-binding potential of properly calcined CCA may have contributed secondarily, depending on its combustion conditions and mineralogical composition.

The combined presence of RHA and CCA appears to provide a synergistic effect that partially suppresses ASR-related expansion, while both ashes can contribute a micro-filler effect that enhances matrix densification. Nevertheless, increased capillary absorption at higher replacements indicates potential durability problems. To ensure the robustness of these findings, future studies should incorporate detailed statistical analysis alongside extended durability testing—such as long-term ASR monitoring, chloride penetration resistance, and freeze-thaw cycling—to validate performance under field conditions and refine optimal replacement levels for practical applications.

3.4. Microstructural Analysis

Microstructural evaluations were performed on all mortar samples from Phase III, namely M0, M5, M10, M15, and M20. These analyses (XRD, TGA, SEM) were conducted to test the hypothesis that rice husk ash (RHA), due to its high amorphous silica content, would accelerate early calcium silicate hydrate (C-S-H) nucleation, while corn cob ash (CCA), with its slower reactivity, would contribute to later-age hydrate formation. Accordingly, XRD and TGA were used to monitor crystalline phase evolution and quantify calcium hydroxide (CH) consumption, while SEM provided visual evidence of morphological refinement and C-S-H gel development. Together, these methods clarify the mechanisms behind the observed trends in durability and mechanical performance.

3.4.1. X-ray Diffraction (XRD)

X-ray diffraction of mortars (M0 to M20) from 10 to 60° 2 θ are given in Figure 5. The aim was to follow crystalline phase evolution (portlandite, quartz, alite, belite, ettringite, calcite) and to infer amorphous C-S-H development from changes in crystalline peak intensities and background intensity. Major crystalline peaks detected in all samples: portlandite at $\approx 18^\circ$, 34.5°, 47.5°, and 48.5° 2 θ ; quartz at $\approx 20.9^\circ$, 26.9°, and 43.5° 2 θ ; alite near $\approx 29^\circ$ 2 θ (see patterns in Figure 5). No new crystalline phases appeared with RHA-CCA replacement; the principal variation among mixes was in relative peak intensities rather than presence/absence of phases.

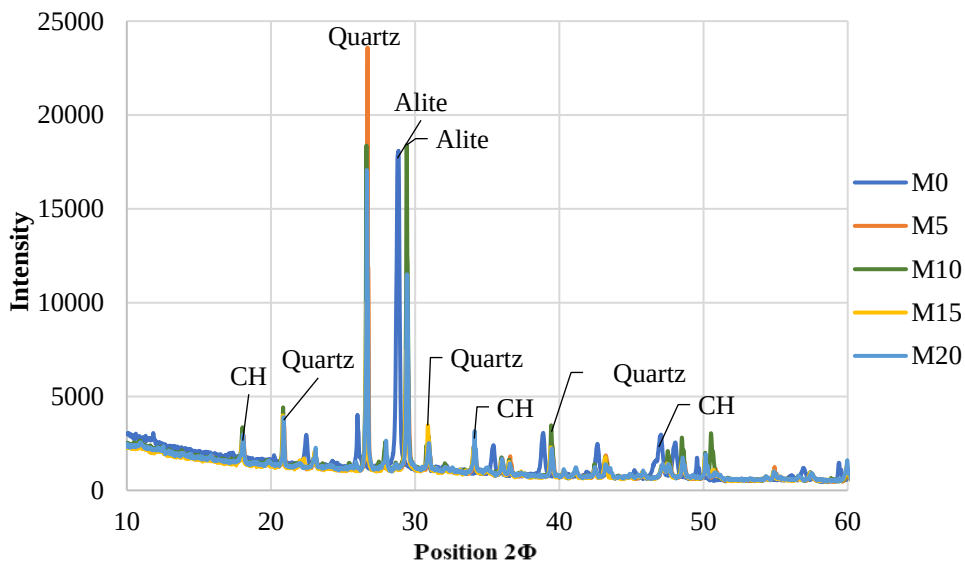


Figure 5. X-ray diffraction for different mortars from 10 to 60° 2 θ .

Figure 6 and Figure 7 show the portlandite ($\text{Ca}(\text{OH})_2$) and quartz (SiO_2) percentages in various mortar specimens according to quantitative analysis of the XRD test. CH peak intensities decreased as the replacement increased, indicating a progressive consumption of free $\text{Ca}(\text{OH})_2$ through pozzolanic reactions. Semi-quantitative XRD values derived from peak

integration are shown in Figure 6: M0 (13.3%), M5 (8.3%), M10 (6.7%), M15 (8.7%), M20 (7.9%). The strongest CH depletion occurs at low-moderate replacements (M5–M10), consistent with active RHA-driven pozzolanic reaction. The partial rebound at M15 is likely due to sampling heterogeneity or incomplete reaction of less-reactive CCA particles.

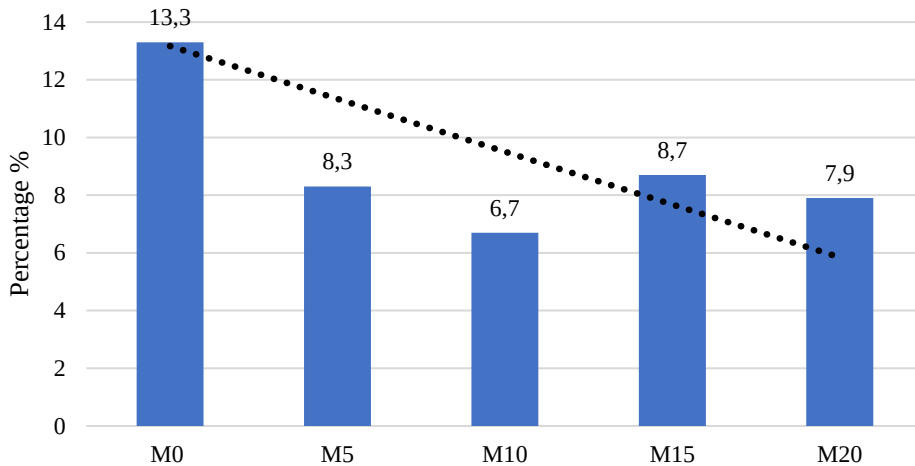


Figure 6. Portlandite ($\text{Ca}(\text{OH})_2$) percent in mortars.

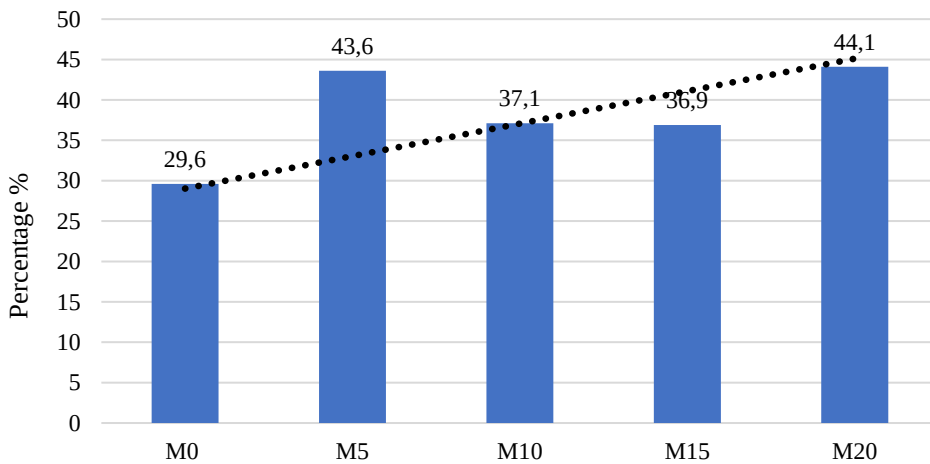


Figure 7. Quartz (SiO_2) percent in mortars.

The crystalline quartz fraction increased with replacement: M0: 29.6%, M5: 43.6%, M10: 37.1%, M15: 36.9%, M20: 44.1% as can be seen in Figure 7. The rise in quartz represents the larger contribution of crystalline or slowly reactive silica from RHA/CCA in the binder fraction. Increased quartz does not impede pozzolanic activity but signals that a portion of silica remains crystalline/inert at the examined curing age.

Although phases such as C–S–H and ettringite are largely amorphous and thus not directly detectable via XRD, the observed changes in crystalline components indirectly support their formation. A modest increase in calcite (CaCO_3) reflections was also observed, consistent with partial carbonation of residual CH or transformation of secondary hydration products during curing/handling.

These findings are consistent with the microstructural and thermal analyses presented in later sections, collectively confirming the pozzolanic activity and its contribution to matrix densification.

3.4.2. Thermogravimetric Analysis (TGA)

The TGA results are presented in Figure 8, and CH quantification is plotted in Figure 9. CH was calculated following the method proposed by Aliabdo et al. (2016), based on the mass loss between 400–600 °C using Equation 1, where W_T is the overall weight reduction percentage between 400 and 600 °C, W_{CH} is the molecular weight of CH (74 g/mol), and W_W is the molecular weight of water (18 g/mol).

$$CH (\%) = W_T * \left(\frac{W_{CH}}{W_W} \right) \quad (1)$$

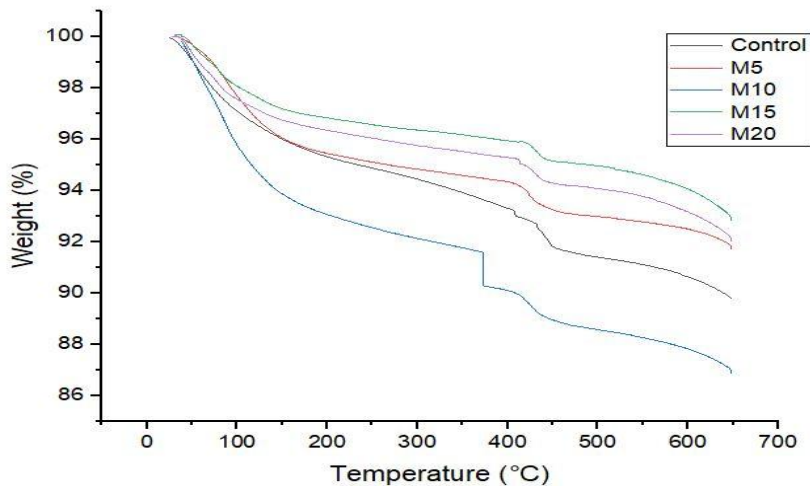


Figure 8. Correlation between the temperature and weight loss.

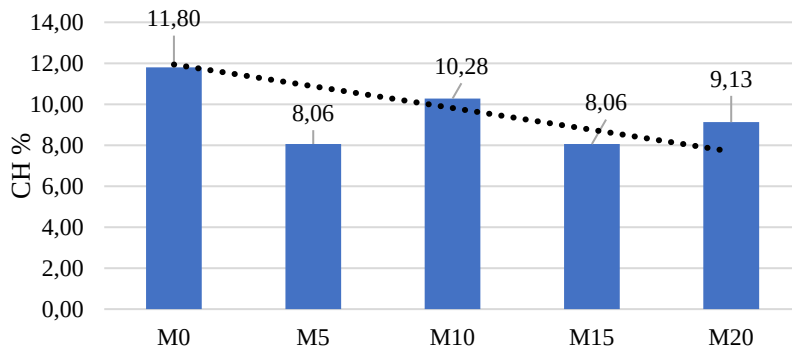


Figure 9. Calcium hydroxide (CH) content in different replacement percentages.

As shown in Figure 9, CH content exhibited a generally decreasing trend with increasing replacement levels, consistent with the expected pozzolanic behavior of CCA and RHA. The control mixture (M0) showed the highest CH content (11.80%), while all modified mixtures demonstrated varying degrees of reduction. These results suggest that while the pozzolanic reaction generally leads to CH reduction, the rate and extent of this reaction may vary depending on the replacement percentage and the interaction between CCA and RHA. The observed CH depletion supports the XRD findings, which indicated diminished portlandite peaks in mixtures with higher additive content. This correlation reinforces the role of pozzolanic materials in promoting secondary C-S-H formation and enhancing the cementitious matrix.

Although M5 and M15 yielded slightly lower CH values than expected relative to M10 and M20, these variations remain within a plausible range and do not disrupt the overall downward trend. Such fluctuations may be attributed to sample heterogeneity or localized differences in ash reactivity, or

incomplete hydration at the time of testing. Nevertheless, the results confirm the broader interpretation of the pozzolanic effectiveness of the RHA-CCA blend and its contribution to matrix densification.

3.4.3. Scanning Electron Microscopy (SEM)

The SEM analyses revealed progressive microstructural refinement across the mortar mixtures with increasing levels of CCA and RHA replacement. The control specimen (M0), shown in Figure 10, exhibited a porous matrix dominated by large, well-defined calcium hydroxide (CH) crystals, often accompanied by microcracks and voids. These features suggest limited pozzolanic activity and potential vulnerability to long-term durability loss.

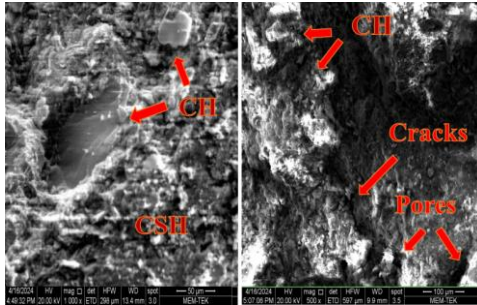


Figure 10. SEM images of the control specimen (M0).

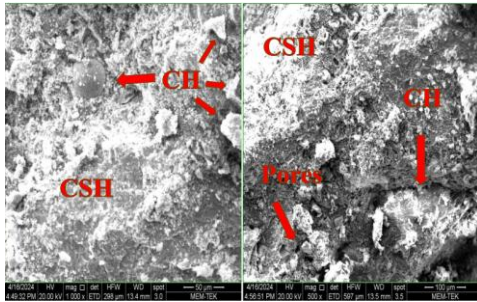


Figure 11. SEM images of M5.

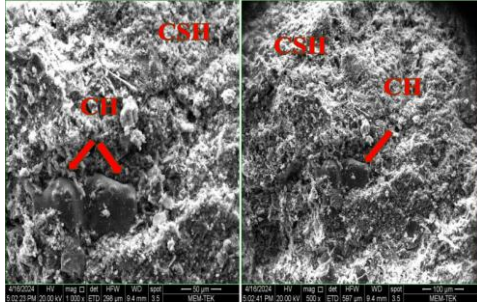


Figure 12. SEM images of M10.

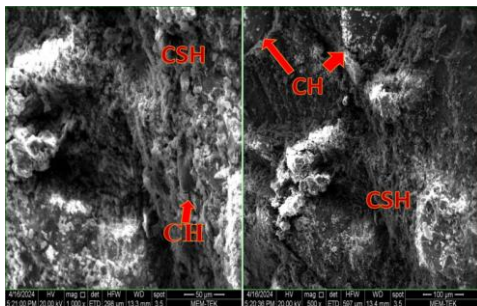


Figure 13. SEM images of M15.

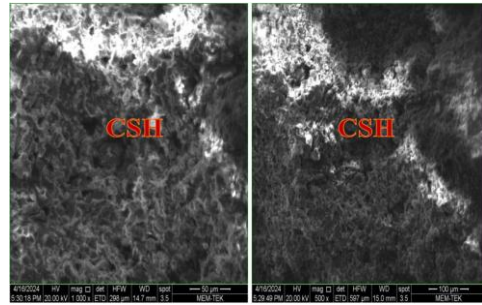


Figure 14. SEM images of M20.

In contrast, the modified mixtures demonstrated enhanced morphological characteristics:

- M5 (Figure 11) showed a slight increase in calcium-silicate-hydrate (C-S-H) gel formation, though CH crystals and microcracks remained visible. This suggests that pozzolanic reactions had initiated but were not yet sufficient to significantly densify the matrix.
- M10 (Figure 12) presented a noticeably denser structure with reduced CH content and more cohesive C-S-H gel networks. The absence of visible cracks suggests improved internal bonding and microstructural stability.
- M15 (Figure 13) exhibited further suppression of CH crystals and increased continuity in the C-S-H phase. The matrix appeared more compact, reflecting ongoing secondary hydration and enhanced particle integration.
- M20 (Figure 14) displayed the most refined morphology, with near-complete elimination of CH crystals and extensive C-S-H coverage. Although some porosity persisted, the overall structure was markedly homogeneous and tightly bonded.

These morphological trends align with the quantitative findings from XRD and TGA analyses, confirming the sustained pozzolanic activity and reduction in free lime content over time. While early-age mechanical strength showed a decreasing trend with higher SCM content, the SEM observations suggest a promising trajectory for long-term durability and matrix densification. The dual contribution of CCA and RHA appears to foster a synergistic refinement of the microstructure, particularly at higher replacement levels.

4. Conclusions

The mechanical and microstructural performance of cement mortars incorporating corn cob ash (CCA) and rice husk ash (RHA) as blended supplementary cementitious materials (SCMs) was investigated in this study. A three-phase experimental program was conducted to assess pozzolanic activity, optimize blend ratios, and evaluate performance across varying replacement levels. Based on the findings, the following conclusions can be drawn:

- **Pozzolanic Reactivity:** RHA exhibited high pozzolanic activity due to its amorphous silica content and industrial-grade fineness, contributing to early-age strength and matrix densification. CCA showed limited reactivity at lower calcination temperatures but reached acceptable pozzolanic activity index (PAI) values when calcined at 700 °C, justifying its inclusion in blended systems.
- **Blend Optimization:** The 15C5R blend (75% CCA + 25% RHA within the replacement fraction) demonstrated the most balanced mechanical performance, combining early-age strength retention with long-term durability. This blend was selected for further evaluation in Phase III.
- **Mechanical Performance:** Mortars with 5–10% total binder replacement using the optimal blend retained 92–97% of the reference (M0) compressive strength at 28 days. Even at 20% replacement, the mix achieved 49.2 MPa, indicating its suitability for general-purpose applications. Flexural strength followed a similar trend, with moderate reductions remaining within acceptable engineering limits.
- **Durability Indicators:** Capillary water absorption increased with higher replacement levels, reflecting a more porous matrix due to reduced cement content. However, ASR expansion values decreased from 0.34% to 0.28%, suggesting partial mitigation through the combined chemical and physical effects of RHA and CCA. While this reduction suggests improved alkali binding matrix densification, the expansions still remain above the acceptable threshold of 0.10%. Therefore, the blended ashes show promise for durability enhancement, but further long-term studies are recommended.
- **Microstructural Evolution:** XRD, TGA, and SEM analyses confirmed progressive consumption of calcium

hydroxide (CH) and increased formation of calcium silicate hydrate (C–S–H) with rising SCM content. SEM images revealed enhanced morphological characteristics, including reduced CH crystals, improved C–S–H continuity, and refined pore structure—particularly in M15 and M20 mixtures.

- **Sustainability Implications:** The use of agricultural waste ashes not only reduces Portland cement consumption and associated CO₂ emissions but also promotes waste valorization and local resource utilization. The synergistic interaction between RHA and CCA can be a good solution for eco-efficient binder design, especially in regions with abundant biomass waste.

In conclusion, blended systems incorporating RHA and CCA—especially at 5–10% replacement levels—can deliver satisfactory mechanical performance and partial durability improvement while contributing to sustainable construction practices. Future research should focus on enhancing mix optimization and durability through controlled particle grading, calcination parameters, and the use of chemical admixtures. In addition, long-term field exposure studies, life-cycle assessments, and comparative evaluations with other agricultural ashes are recommended to validate real-world performance and broaden the applicability of these waste-derived binders.

Declaration of Ethical Standards

The authors declare that they comply with all ethical standards.

Credit Authorship Contribution Statement

Author 1: Research, Methodology / Study design, Writing – original draft, Data curation
Author 2: Conceptualization, Methodology, Supervision, Project administration, Funding acquisition, Visualization, Writing – review and editing

Declaration of Competitive Interests

The authors have no conflict of interest to declare regarding the content of this article.

Data Availability

All data generated or analyzed during this study are included in this published article. Data will be made available if deemed necessary.

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5. References

- Abellan-García, J. (2023). Effect of rice husk ash as partial replacement of ordinary Portland cement in ultra-high-performance glass concrete. *European Journal of Environmental and Civil Engineering*, 28(3), 661–683. <https://doi.org/10.1080/19648189.2023.2219722>
- Abubakar, A., Mohammed, A., Duna, S., & Samson, D. (2016). Mechanical properties of concrete containing corn cob ash. *International Journal of Scientific Research and Engineering Studies (IJSRES)*, 3(6), 47–51.

- Adebisi, O., Taiwo, A. M., Julius, O. B., & Ebenezer, A. (2019). *Partial replacement of cement with corn cob ash – a review*. Retrieved September 13, 2025, from https://www.globalscientificjournal.com/researchpaper/PARTIAL_REPLACEMENT_OF_CEMENT_WITH_CORN_COB_ASH_A_REVIEW.pdf
- Aliabdo, A. A., Abd Elmoaty, A. E. M., & Aboshama, A. Y. (2016). Utilization of waste glass powder in the production of cement and concrete. *Construction and Building Materials*, 124, 866–877. <https://doi.org/10.1016/j.conbuildmat.2016.08.016>
- ASTM International. (2020). ASTM C185: Standard test method for air content of hydraulic cement mortar. West Conshohocken, PA: ASTM International.
- ASTM International. (2020). ASTM C1585: Standard test method for measurement of rate of absorption of water by hydraulic-cement concretes. West Conshohocken, PA: ASTM International.
- ASTM International. (2022). ASTM C618: Standard specification for coal fly ash and raw or calcined natural pozzolan for use in concrete. West Conshohocken, PA: ASTM International.
- ASTM International. (2025). ASTM C311: Standard test methods for sampling and testing coal ash or natural pozzolans for use in concrete. West Conshohocken, PA: ASTM International.
- ASTM International. (2025). ASTM C1567: Standard test method for determining the potential alkali-silica reactivity of combinations of cementitious materials and aggregate (accelerated mortar-bar method). West Conshohocken, PA: ASTM International.
- Bahri, S., Mahmud, H. B., Shafigh, P., & Majuar, E. (2019). Mechanical and durability properties of high strength high performance concrete incorporating rice husk ash. *IOP Conference Series: Materials Science and Engineering*, 536, 012028. <https://doi.org/10.1088/1757-899X/536/1/012028>
- British Standards Institution. (1999). BS EN 1015-3: Methods of test for mortar for masonry – Part 3: Determination of consistence of fresh mortar (by flow table). London: BSI.
- British Standards Institution. (2002). BS EN 1015-18: Methods of test for mortar for masonry – Part 18: Determination of water absorption coefficient due to capillary action of hardened mortar. London: BSI.
- British Standards Institution. (2011). BS EN 197-1: Cement – Part 1: Composition, specifications and conformity criteria for common cements. London: BSI.
- British Standards Institution. (2012). BS EN 450-1: Fly ash for concrete – Part 1: Definition, specifications and conformity criteria. London: BSI.
- British Standards Institution. (2012). BS EN 934-2: Admixtures for concrete, mortar and grout – Part 2: Concrete admixtures – Definitions, requirements, conformity and marking. London: BSI.
- British Standards Institution. (2016). BS EN 196-1: Methods of testing cement – Part 1: Determination of strength. London: BSI.
- Ekincioglu, O., Gurgun, A.P., Engin, Y., Tarhan, M., Kumbaracibasi, S., (2013). Approaches for sustainable cement production- A case study from Turkey, *Energy and Buildings*, 66, 136-142. <https://doi.org/10.1016/j.enbuild.2013.07.006>
- Ghosh, S. (Ed.). (2018). *Sustainable construction materials*. Amsterdam: Elsevier.
- Jayaraman, A., Vasudevan, M., Aravind, R., Keerthika, M., & Hari Prakash, J. (2023). Optimization of partial replacement of cement by silica fume and rice husk ash for sustainable concrete. *Materials Today: Proceedings*. <https://doi.org/10.1016/j.matpr.2023.05.598>

- Kamau, J., Ahmed, A. Hirst, P., & Kangwa, J., (2016). Suitability of corncob ash as a supplementary cementitious material. *International Journal of Materials Science and Engineering*, 4(4), 215-228. <https://doi.org/10.17706/ijmse.2016.4.4.215-228>
- Kumari, S., Chander, D., & Walia, R. (2018). Durability and strength analysis of concrete by partial replacement of cement with corn cob ash and rice husk ash. *International Journal of Research in Advent Technology*, 6(7), 1593-1601.
- Le, H. T., Nguyen, S. T., & Ludwig, H. M. (2014). A study on high performance fine-grained concrete containing rice husk ash. *International Journal of Concrete Structures and Materials*, 8(4), 301-307. <https://doi.org/10.1007/s40069-014-0078-z>
- Mehta, P. K., & Monteiro, P. J. M. (2014). *Concrete: Microstructure, properties, and materials* (4th ed.). New York: McGraw-Hill Education.
- Noori, A. N., Salman, H. K., Numan, H. A., & Al-Dlaemee, S. A. H. (2023). Effect of Iraqi's rice husk ash on the mechanical properties of cement mortar: An experimental investigation. *AIP Conference Proceedings*, 2787(1), 080042. <https://doi.org/10.1063/5.0149200>
- Olafusi, O. S., Kupolati, W. K., Sadiku, E. R., Snyman, J., and Ndambuki, J. M. (2018). Characterization of corncob ash (CCA) as a pozzolanic material. *International Journal of Civil Engineering and Technology (IJCIET)*, 9(12), 1016-1024.
- Olaleye, O. T. (2022). Performance of corncob ash pozzolan in concrete samples. *Quest Journals: Journal of Architecture and Civil Engineering*, 7(12), 15-20. ISSN (Online): 2321-8193.
- Olaniyi, A. (2022). Strength characteristics of rice husk and corn cob ash blended cement concrete. *Sustainability in Environment*, 7(1), 102-110. <https://doi.org/10.22158/se.v7n1p102>
- Siddika, A., Mamun, M. A., Alyousef, R., and Mohammadhosseini, H. (2021). State-of-the-art-review on rice husk ash: A supplementary cementitious material in concrete. *Journal of King Saud University – Engineering Sciences*, 32(5), 294-307. <https://doi.org/10.1016/j.jksues.2020.10.006>
- Siddique, R., & Khan, M. I. (2011). *Supplementary cementing materials*. Dordrecht: Springer.
- Singh, B. (2018). Rice husk ash. In R. Siddique & M. I. Khan (Eds.), *Waste and supplementary cementitious materials in concrete: characterization, properties and applications* (pp. 417-460). Amsterdam: Elsevier. <https://doi.org/10.1016/B978-0-08-102156-9.00013-4>
- Tiseo, I. (2025). Cement industry emissions worldwide – Statistics & Facts. Statista. Retrieved October 26, 2025, from <https://www.statista.com/topics/11056/cement-industry-emissions-worldwide/>